

4(a)	work (done) = force $\times$ displacement	M1
	(force = mg and distance = Δh)	A1
	(so) work (done) = $mg\Delta h$ <u>and</u> work = $\Delta E_{(P)}$ (so $\Delta E_{(P)} = mg\Delta h$)	
4(b)	gravitational potential (energy) to heat/thermal (energy)	B1
4(c)(i)	$P = mg(\Delta)h / (\Delta)t$ or Fv	C1
	$P = (0.60 \times 9.81 \times 1.4) / 4.0$ or $0.60 \times 9.81 \times (1.4 / 4.0)$	A1
	= 2.1 W	
4(c)(ii)	$P = I^2R$ or IV or V^2 / R	C1
	= $0.09^2 \times 47$ or 0.09×4.23 or $4.23^2 / 47$	A1
	= 0.38 W	
4(c)(iii)	efficiency = $P_{out} / P_{in} (\times 100)$ or $E_{out} / E_{in} (\times 100)$	C1
	= $0.38 / 2.1 (\times 100)$ or $0.38 \times 4.0 / 2.1 \times 4.0 (\times 100)$	A1
	= 0.18 or 18%	